ColonCrafter: A Depth Estimation Model for Colonoscopy Videos Using Diffusion Priors

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Motivation

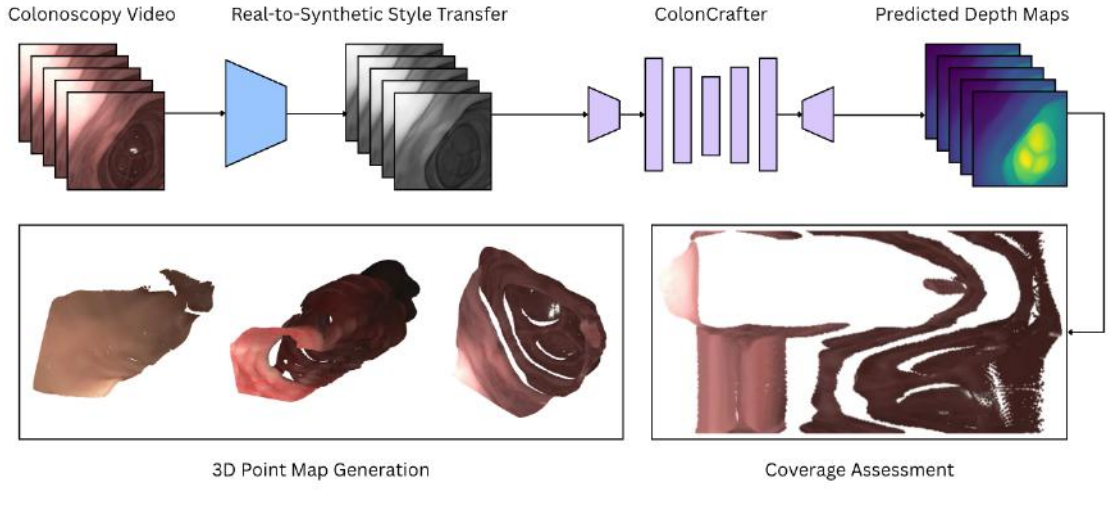
1. Colorectal cancer (CRC) is the second-leading cause of cancer death
2. Colonoscopy is the gold-standard for CRC screening
3. **3D understanding of the colon** could empower clinical decision-making

Challenges

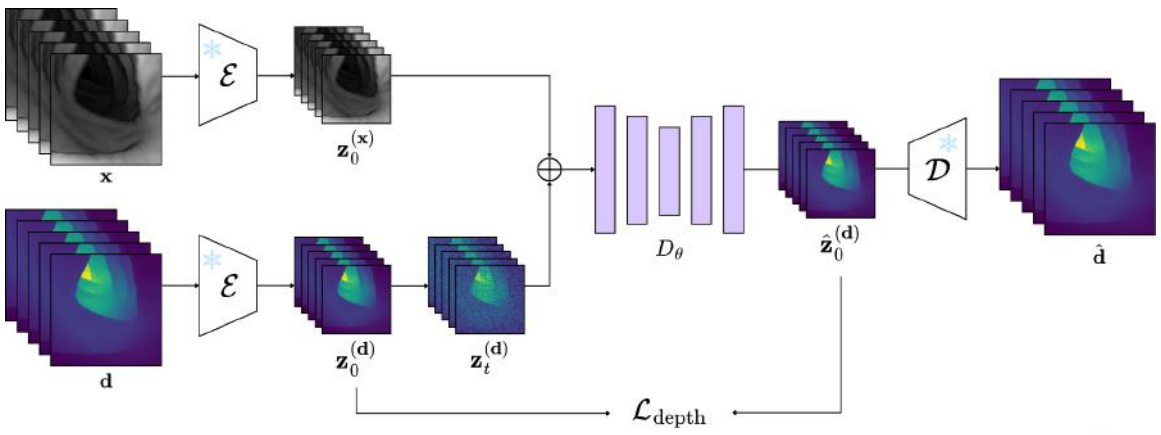
1. Endoscopy videos violate assumptions of standard SLAM: specular highlights, deformable surfaces, lighting changes
2. Existing depth estimation models struggle with **temporal consistency**

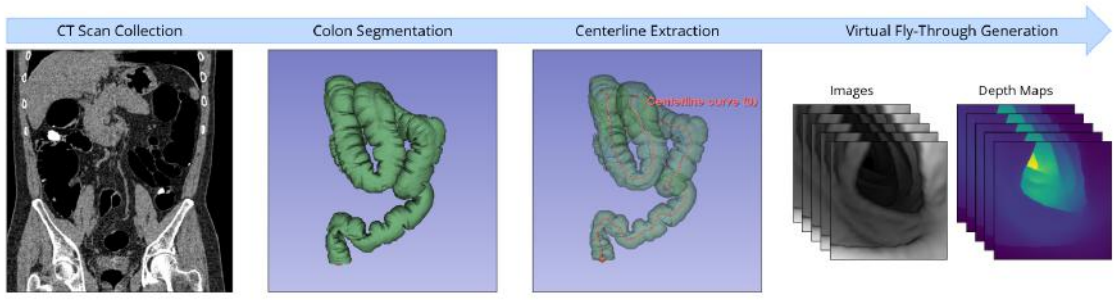
Contributions

1. **Diffusion-based depth estimation:** generates temporally consistent depth maps
2. **Style transfer technique:** maps real colonoscopy videos to the synthetic domain while preserving geometry
3. **Clinically relevant applications:** 3D reconstruction, lesion localization, surface coverage assessment

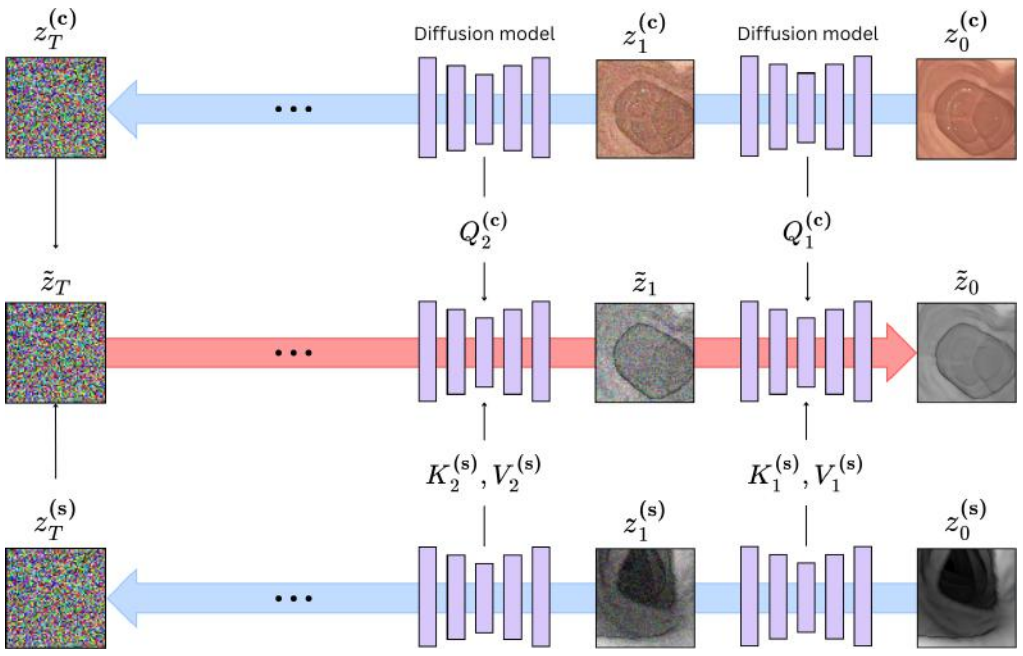


Conditional Video Diffusion for Depth Estimation

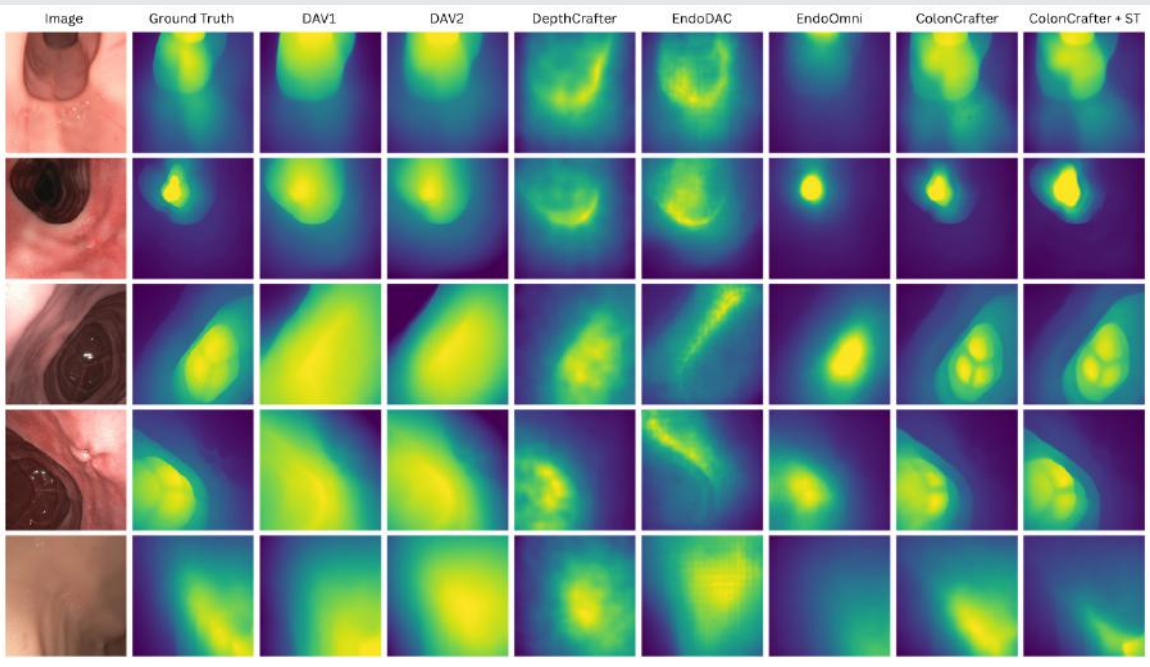




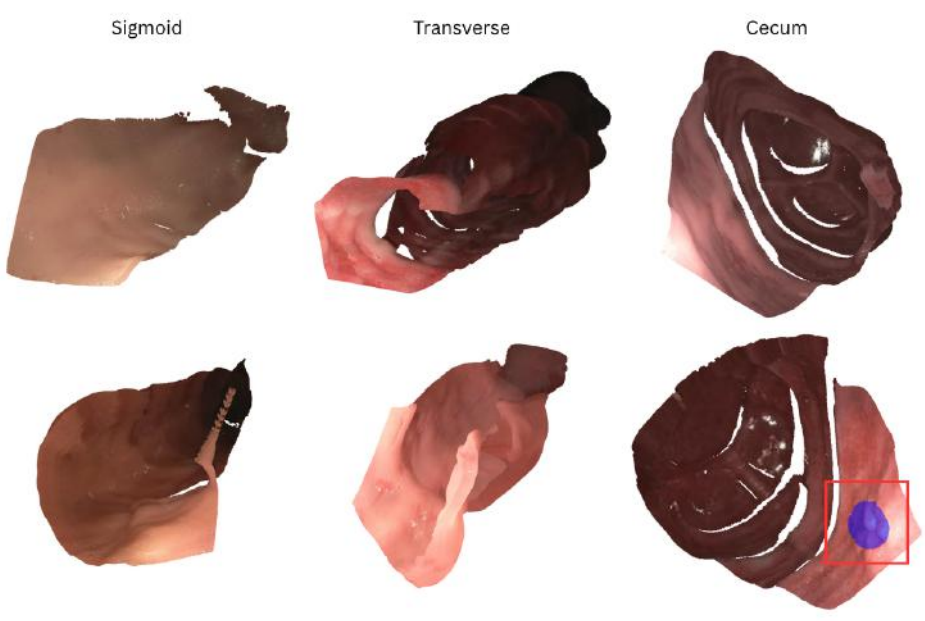
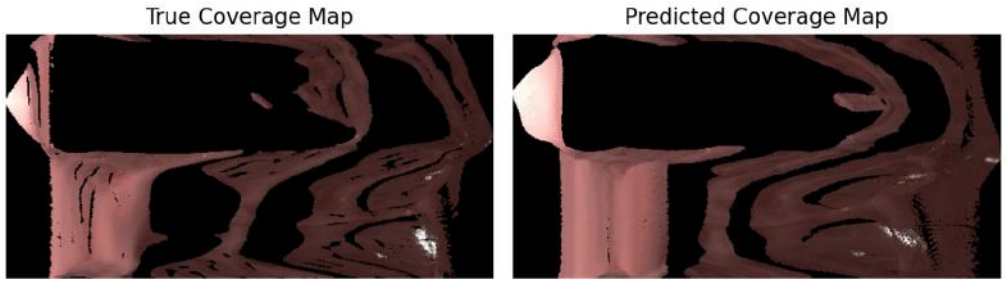
Training-Free Domain Adaptation



Evaluation on Phantom Data



Model	$\delta_1 \uparrow$	AbsRel $\downarrow$	SqRel $\downarrow$	RMSE (mm) $\downarrow$
Depth-Anything-V1[43]	0.55 (0.45, 0.66)	0.28 (0.21, 0.34)	3.58 (1.83, 5.50)	10.08 (7.08, 13.24)
Depth-Anything-V2[44]	0.61 (0.52, 0.71)	0.24 (0.19, 0.28)	2.34 (1.55, 3.09)	8.20 (6.33, 9.87)
DepthCrafter[31]	0.59 (0.52, 0.65)	0.22 (0.19, 0.26)	2.65 (2.07, 3.14)	10.51 (9.09, 11.70)
EndoDAC[45]	0.50 (0.42, 0.61)	0.27 (0.22, 0.33)	4.45 (2.64, 6.63)	13.91 (10.45, 17.42)
EndoSfM-Learner[9]	0.56 (0.49, 0.64)	0.24 (0.21, 0.29)	3.49 (2.42, 4.68)	12.34 (9.81, 15.08)
EndoOmni[†46]	0.77 (0.72, 0.81)	0.15 (0.14, 0.16)	1.15 (0.88, 1.43)	6.91 (5.54, 8.20)
ColonCrafter	0.77 (0.66, 0.83)	0.16 (0.13, 0.20)	1.17 (0.81, 1.61)	6.42 (5.09, 7.62)
ColonCrafter + ST	0.79 (0.70, 0.86)	0.15 (0.12, 0.18)	1.09 (0.73, 1.45)	6.21 (5.01, 7.18)

ST: style transfer; †: Not strictly zero-shot (partially trained on C3VD).

Downstream Applications